Statement of Purpose

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Ph.D. in Computer and Information Science and Engineering

An intelligent software system can produce a convincing account of its work without delivering the result a user requested. I want to study the mechanisms that close this gap: how systems preserve an intended goal, recover from partial failures, and provide evidence that their actions produced the correct outcome. My research interests lie in reliable intelligent software systems, with an emphasis on software testing, program analysis, and long-running agent-based workflows. I am applying to Syracuse University's Ph.D. in Computer and Information Science and Engineering to develop these interests into rigorous research grounded in working systems and reproducible experiments.

My preparation combines a Bachelor of Science in Computer Science from the University of British Columbia's Okanagan campus with my ongoing Master of Computer Science at the University of Melbourne. The master's program has combined coursework with a research year. Under Professor Richard Sinnott's supervision, my research concerns Android privacy evidence. Through Privacy Lens, I have worked on an offline, evidence-first research prototype and a reproducibility artifact that separates directly observed behavior from interpretations requiring further review. This experience has made evidence provenance, evaluation design, and the limits of automated interpretation central to how I approach software research.

Alongside my degree, I have pursued independent systems projects addressing persistent execution and multi-agent coordination. DS-Hns focuses on durable task state, restart and recovery behavior, and bounded execution of coding tasks. Codex Boss explores multi-agent orchestration and evidence-based acceptance. I subsequently integrated these lines of work in Utopia, whose demonstrated scope includes two Windows execution workers with Android and web control interfaces. Across these projects, my role has centered on architecture, AI-assisted implementation and integration, and reviewing whether the available evidence supports a completion claim. I maintain versioned development records so that implementation changes, test observations, and unresolved failures remain distinguishable.

These projects have given me concrete questions rather than only a general interest in AI. A task-status transition, a successful tool response, and a correct final application state are different observations. In DS-Hns interruption experiments, recovering committed progress could avoid replay at a saved boundary, while an effect-before-checkpoint gap could still repeat an action. A failure record can support debugging, but it becomes more useful for research when it can be reproduced under a controlled setup and compared with an alternative design. I want to move from solving such problems within my own implementations to understanding which mechanisms generalize across systems and which assumptions limit them.

One direction I would like to investigate is outcome-grounded testing for agent-based software. I would begin with a small collection of benign tasks whose outcomes can be independently checked, such as a repository modification or an update to a local application's state. The study would compare task self-reports and log-based assertions with checks against those outcomes, then introduce controlled interruptions, stale state, and inconsistent tool responses. Useful measures would include false-completion rate, recovery correctness, repeated effects, and the human effort required to diagnose a failure. This would provide a tractable starting point for studying when additional checks improve reliability and when they create excessive cost or misleading confidence.

Another direction concerns preserving constraints during recovery and replanning. An agent may need to change its implementation strategy without changing the user's goal or exceeding the permissions granted for the task. I am interested in how explicit task contracts, program analysis, and runtime checks could help detect that divergence. I would like to strengthen my foundations in analysis and automated reasoning while investigating which properties can be checked effectively in real software. I would evaluate a focused mechanism through baselines and ablations, using the framework as an experimental platform.

Syracuse offers a compelling methodological fit. Professor Endadul Hoque's work on context-sensitive fuzzing, runtime verification, and LLM-driven security analysis connects closely to my interest in exposing failures systematically and testing whether generated behavior follows a specification. Professor Kristopher Micinski's work on scalable logical reasoning and program analysis offers a basis for investigating how task constraints can be expressed and checked beyond observed test runs. I am also interested in Professor Yuzhe Tang's research on software and distributed-systems security, including its connections to AI agents. Together, these areas would allow me to develop a stronger relationship between empirical evaluation, analysis, and dependable systems design.

Wearable and ubiquitous computing remain motivating application settings for me because they make interrupted connectivity, cross-device state, and user control tangible. I would welcome refining the question with a supervisor and working within an established research platform where that produces a stronger study. I want doctoral training to sharpen my ability to formulate useful questions, design convincing evaluations, and communicate contributions that others can inspect and build upon. I would bring sustained systems-building experience and a practical focus on whether intelligent software actually works as intended.